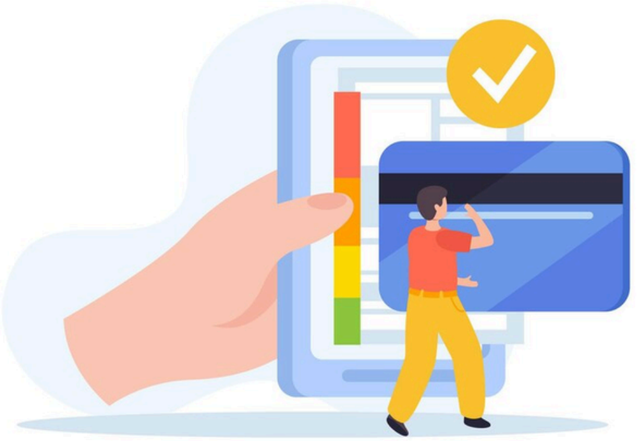
Financial Inclusion

PsyCred

AN ALTERNATIVE CREDIT SCORING SYSTEM



Team:

CreditIQ Crew

PROBLEM STATEMENT

- In countries like India, financial inclusion remains a challenge, particularly for low-income individuals lacking access to traditional banking.
- Credit scores, pivotal in accessing loans and financial services, are often inaccessible due to limited credit history.
- To remedy this, alternative credit scoring methods utilizing AI could leverage unconventional data, fostering greater financial inclusion for marginalized groups.

STATISTICS

SOURCE: Financial Express, Forbes

94.5% of Indians have not been introduced to credit cards

70%

of the rural poor do not have a bank account.

87%

of the rural poor do not have access to credit from a formal source.

17.8%

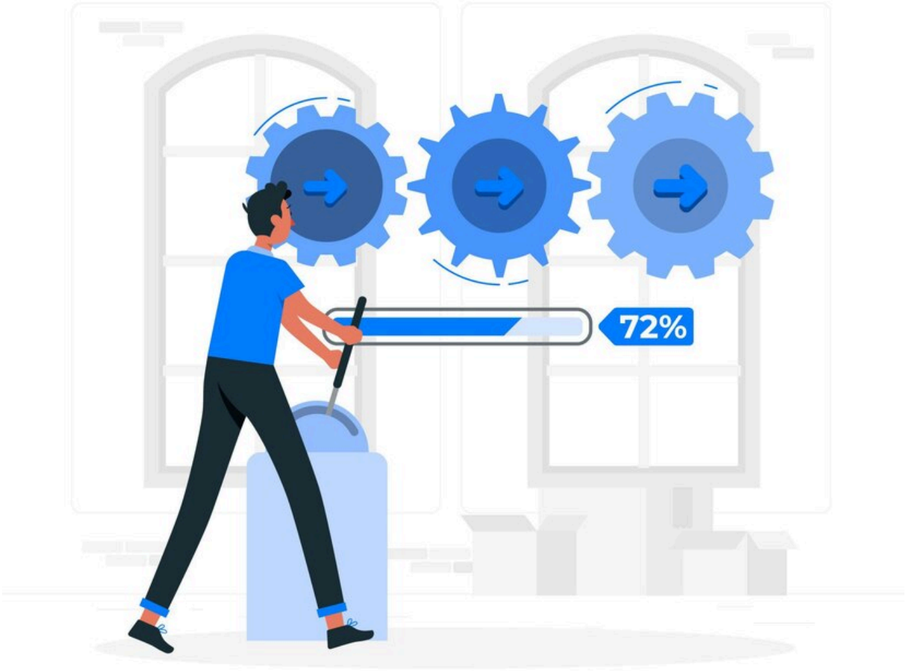
of rural households are indebted to institutional credit agencies, compared to **14.5%** of urban households.

10.2%

of rural households are indebted to non-institutional credit agencies, compared to **4.9%** of urban households.

STAKEHOLDERS

- Consumers / Borrowers
- Banking Institutions



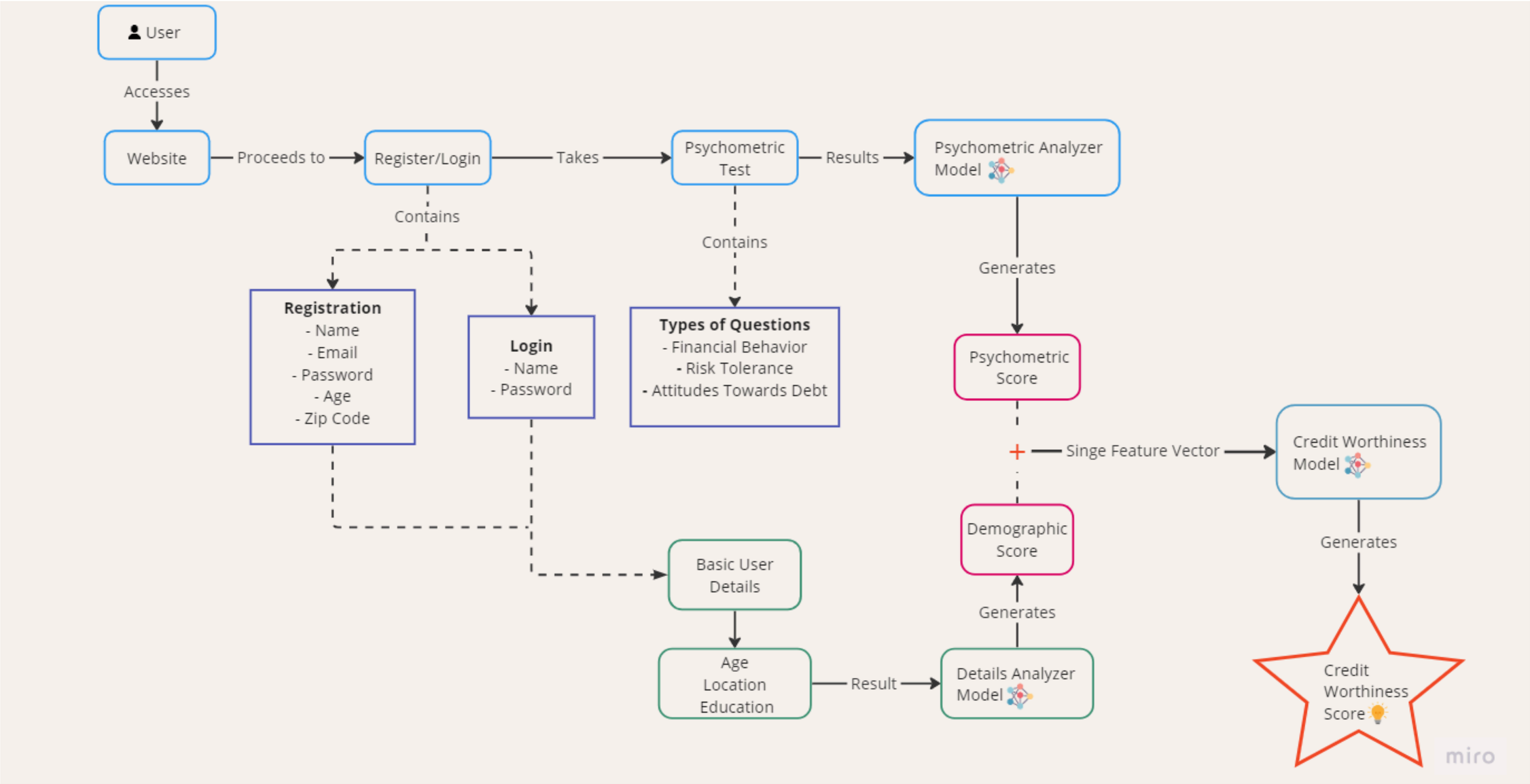
EXISITING SOLUTIONS

- COSMO - Psychometric-test based credit assessment tool
- Microfinance Institutions (MFIs)

But what they DON’T offer

- A transparent and unbiased credit scoring system.
- A scoring system that takes into consideration rural conditions.
- A system that caters to the needs of people who don't have access to financial services.

PROPOSED SOLUTION



PROPOSED SOLUTION

Solution Approach

- Designing a comprehensive psychometric test for assessing creditworthiness.
- Consideration of factors like education and employment history as stability indicators - **Demographic Score**.

Psychometric Data → **Psychometric Score**

User Details → **Demographic Score**

- Combining the Psychometric Score and Demographic Score to define the overall Credit score.

Unique Selling Proposition

- Our solution offers personalized credit assessments powered by advanced psychometric testing and data analysis.
- Unlike traditional methods, we provide insights beyond financial metrics, delivering nuanced evaluations.
- By considering demographic traits, we offer tailored assessments for informed lending decisions, enhancing financial access.

Tech Stack

Flask (Backend Development)

Vanilla JS (Front-end Development)

SQLite (Database)

PyTorch for ML Model

PLAN OF ACTION

Design a Psychometric test

Create the frontend

Use Dataset 1 to train Test Analyser Model

Use Dataset 2 to prepare the Demographic Analyser model

Train the Credit worthiness model

Set up Backend and APIs

Test the Final Prototype

DEMOGRAPHIC SCORE

We had to engineer a placeholder formula based on demographic information

Our formula took a lot of criteria into account including but not limited to:

- a well educated person would always have a higher score
- an older person would have a comparatively lower score
- a person from a rapidly growing economy would have an extremely high score

DEMOGRAPHIC SCORE

The formula based on (E)ducation, (L)iterature rate, (A)ge, (G)rowth of economy

$$S = max\left(0, 1 + \frac{E}{10} + \frac{L - 30}{50} - \frac{A}{25}\right)\frac{max\left(5, G\right)}{30}$$

$$D = 2\sigma\left(S\right) - 1 \qquad \sigma\left(x\right) = \frac{1}{1 + e^{-x}}$$

SYNTHETIC DATASET

```
(22, 'H', 'Patiala', 1895686, 19.62, 891, 75.28) 0.53
(26, 'P', 'Nagaur', 3307743, 19.2, 950, 62.8) 0.68
(33, 'H', 'Gajapati', 577817, 11.37, 1043, 53.49) 0.18
(52, 'M', 'Gwalior', 2032036, 24.5, 864, 76.65) 0.50
(36, 'H', 'Akola', 1813906, 11.27, 946, 88.05) 0.28
(40, 'B', 'East Godavari', 5154296, 5.16, 1006, 70.99) 0.10
(18, 'M', 'Krishnagiri', 1879809, 20.41, 958, 71.46) 0.71
(37, 'P', 'Thane', 11060148, 36.01, 886, 84.53) 0.92
(40, 'M', 'Coimbatore', 3458045, 18.56, 1000, 83.98) 0.55
(41, 'H', 'Nagpur', 4653570, 14.4, 951, 88.39) 0.31
(50, 'H', 'Gurgaon', 1514432, 73.96, 854, 84.7) 0.80
(56, 'H', 'Jajapur', 1827192, 12.49, 973, 80.13) 0.12
(40, 'B', 'Firozpur', 2029074, 16.21, 893, 68.92) 0.31
(42, 'B', 'Kamrup', 1517542, 15.69, 949, 75.55) 0.31
(27, 'M', 'Madurai', 3038252, 17.84, 990, 83.45) 0.63
(43, 'M', 'Bikaner', 2363937, 41.19, 905, 65.13) 0.77
(52, 'H', 'Sehore', 1311332, 21.54, 918, 70.06) 0.18
(41, 'M', 'Balaghat', 1701698, 13.6, 1021, 77.09) 0.39
(30, 'M', 'Davanagere', 1945497, 8.63, 972, 75.74) 0.31
(41, 'B', 'Vadodara', 4165626, 14.38, 934, 78.92) 0.31
(35, 'P', 'Srinagar', 1236829, 20.35, 900, 69.41) 0.67
(27, 'B', 'Vadodara', 4165626, 14.38, 934, 78.92) 0.43
(28, 'B', 'West Nimar', 1873046, 22.85, 965, 62.7) 0.53
(42, 'P', 'Barddhaman', 7717563, 11.92, 945, 76.21) 0.42
(32, 'H', 'Dhanbad', 2684487, 11.99, 909, 74.52) 0.27
(45, 'P', 'Visakhapatnam', 4290589, 11.96, 1006, 66.91) 0.37
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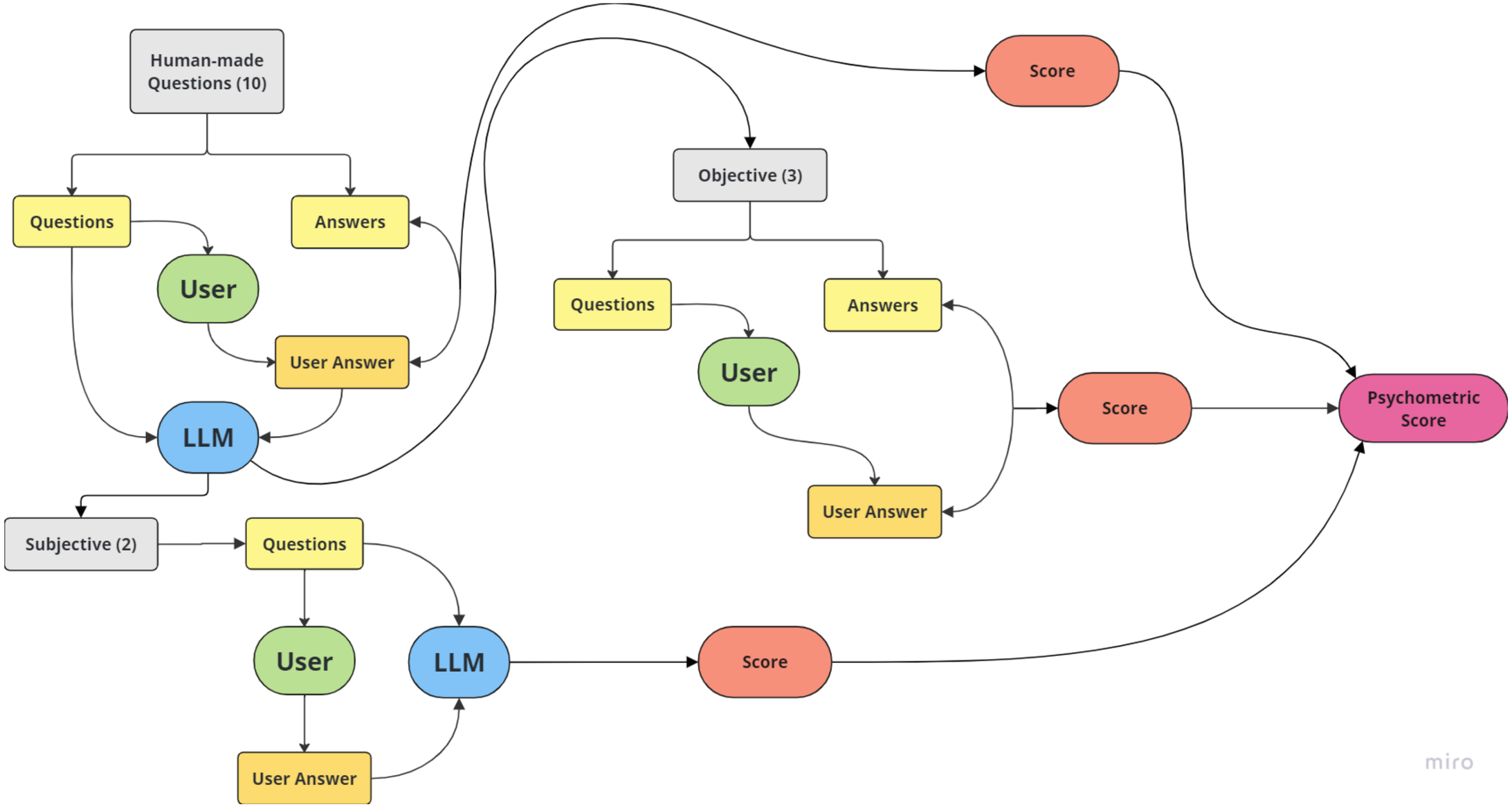
PSYCHOMETRIC SCORE



Thank you!

FINANCIAL INCLUSION: UNLOCKING OPPORTUNITIES FOR THE UNBANKED

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